Quadratic stabilization of the canonical Foulkes–Howe map

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Abstract

For every finite-dimensional complex vector space V, we prove that the canonical Foulkes–Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever $a \geq 2$ and $b \geq a(a-1)$. This gives a quadratic stabilization bound independent of $\dim V$.

1 The result

1.1 Setting

Throughout, V is a finite-dimensional complex vector space, $\mathrm{Sym}\, V = \bigoplus_{j\geq 0} \mathrm{Sym}^j V$ is its symmetric algebra, and a product of vectors always means a product in $\mathrm{Sym}\, V$. Fix an integer $a \geq 1$; $\mathfrak{S}_a$ is the symmetric group on a letters.

Averaging. For a complex vector space W, the group $\mathfrak{S}_a$ acts on $W^{\otimes a}$ by permuting the tensor factors; $(W^{\otimes a})^{\mathfrak{S}_a}$ denotes the invariant tensors. The *averaging operator* is the linear map $\mathrm{Av}\colon W^{\otimes a} \to W^{\otimes a}$ given on simple tensors by

$$\mathrm{Av}(w_1 \otimes \cdots \otimes w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} w_{\pi(1)} \otimes \cdots \otimes w_{\pi(a)}. \tag{1}$$

We use three elementary facts.

(Av1) Av takes values in $(W^{\otimes a})^{\mathfrak{S}_a}$ and fixes every invariant tensor. Hence Av is a projection onto $(W^{\otimes a})^{\mathfrak{S}_a}$, and the elements $\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)$ span $(W^{\otimes a})^{\mathfrak{S}_a}$.

(Av2) The right side of (1) does not change when the w_i are permuted, so Av induces a linear map $\overline{\mathrm{Av}}\colon \mathrm{Sym}^a W \to (W^{\otimes a})^{\mathfrak{S}_a}$, $w_1 \cdots w_a \mapsto \mathrm{Av}(w_1 \otimes \cdots \otimes w_a)$.

(Av3) $\overline{\mathrm{Av}}$ is an isomorphism; its inverse is the restriction to invariant tensors of the quotient map $q\colon W^{\otimes a} \to \mathrm{Sym}^a W$. Indeed, q is unchanged by permuting factors, so $q(\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)) = w_1 \cdots w_a$, i.e. $q \circ \overline{\mathrm{Av}} = \mathrm{id}$. Conversely, $\overline{\mathrm{Av}} \circ q = \mathrm{Av}$ on all of $W^{\otimes a}$, because both sides are linear and agree on simple tensors by the definition in (Av2); so for an invariant τ, $\overline{\mathrm{Av}}(q(\tau)) = \mathrm{Av}(\tau) = \tau$ by (Av1).

The algebra R. The tensor power $(\mathrm{Sym}\, V)^{\otimes a}$ is a commutative algebra under factorwise multiplication, $(s_1 \otimes \cdots \otimes s_a)(s'_1 \otimes \cdots \otimes s'_a) = s_1 s'_1 \otimes \cdots \otimes s_a s'_a$, and $\mathfrak{S}_a$ acts on it by algebra automorphisms. For $j \geq 0$ put

$$R_j = \big((\mathrm{Sym}^j V)^{\otimes a}\big)^{\mathfrak{S}_a}, \qquad R = \bigoplus_{j\geq 0} R_j \subseteq (\mathrm{Sym}\, V)^{\otimes a}.$$

Products of invariants are invariant, and $R_i R_j \subseteq R_{i+j}$, so R is a graded commutative algebra. By (Av3), applied to $W = V$ and to $W = \operatorname{Sym}^b V$,

$$R_1 \cong \operatorname{Sym}^a V, \qquad R_b \cong \operatorname{Sym}^a(\operatorname{Sym}^b V).$$

The map. Let $b \geq 0$. Since R is commutative, $(z_1, \ldots, z_b) \mapsto z_1 \cdots z_b$ is a symmetric b-linear map $R_1^b \to R_b$, hence induces a linear map $\operatorname{Sym}^b R_1 \to R_b$. Under the two identifications above this is the *canonical Foulkes–Howe map*

$$\mu_{a,b,V} : \operatorname{Sym}^b(\operatorname{Sym}^a V) \longrightarrow \operatorname{Sym}^a(\operatorname{Sym}^b V). \tag{2}$$

Let $A \subseteq R$ be the subalgebra generated by R_1 and $A_b = A \cap R_b$ its degree-b part, i.e. the span of all products of b elements of R_1. By construction, $\operatorname{im} \mu_{a,b,V}$ corresponds to A_b. Concretely (see Lemma 3): $\mu_{a,b,V}$ sends the product in $\operatorname{Sym}^b(\operatorname{Sym}^a V)$ of the b elements $u_1^a, \ldots, u_b^a \in \operatorname{Sym}^a V$ to the a-th power, formed in $\operatorname{Sym}^a(\operatorname{Sym}^b V)$, of the single element $u_1 \cdots u_b \in \operatorname{Sym}^b V$.

1.2 Main theorem

Theorem 1. *For every integer $a \geq 2$, every integer $b \geq a(a-1)$, and every finite-dimensional complex vector space V, the map $\mu_{a,b,V}$ in (2) is surjective. Equivalently, $A_b = R_b$.*

For $a = 1$ we have $R = \operatorname{Sym} V$ and $\mu_{1,b,V}$ is the identity in every degree. The bound $a(a-1)$ does not involve $\dim V$. It is a sufficient bound; it is not claimed to be sharp.

1.3 Proof at a glance

Key idea. Surjectivity is dualized into a vanishing statement about multilinear forms, and the vanishing is proved by the pattern *"differentiate into a known zero, then cancel the derivatives"*. The cancellation is possible because a directional derivative $D_{Y,U} = \sum_k Y_k \, \partial/\partial U_k$ is *injective* on the space of polynomials that are homogeneous of degree p in the vector variable U and homogeneous of degree q in the vector variable Y, whenever $p > q$ (Lemma 6).

Steps.

1. *Dualize (Lemma 4).* Call $u_1 \cdots u_b \in \operatorname{Sym}^b V$ a *product of b vectors*. Then $\mu_{a,b,V}$ is surjective if and only if the only symmetric a-linear form T on $\operatorname{Sym}^b V$ with $T(P, \ldots, P) = 0$ for all products P of b vectors is $T = 0$.

2. *Vanishing (Proposition 9).* This is proved for $b \geq a(a-1)$ by induction on a. Write $r = a - 1$ and $m = b - r$; the hypothesis $b \geq a(a-1)$ is exactly $m \geq r^2$. The goal is $T(v_1^b, \ldots, v_r^b, t^b) = 0$ for all vectors, which suffices because b-th powers span $\operatorname{Sym}^b V$ (Lemma 2).

   - *Step 1 (factoring over $\mathbb{C}$).* For a product Q of m vectors, $(x^r + ct^r)Q$ is a product of b vectors for every $c \in \mathbb{C}$. The coefficient of c in the hypothesis gives $T(x^r Q, \ldots, x^r Q, t^r Q) = 0$. Here every slot contains the same Q.
   - *Step 2 (first differential sequence).* To replace the last slot $t^r Q$ by t^b, start from $H_x = T(x^r Q, \ldots, x^r Q, t^b)$ with $Q = y_1 \cdots y_m$ and apply $D_{y_1,t}, \ldots, D_{y_m,t}$. Each operator trades one factor t of the last slot for one y_i, and the end result is a multiple of the Step 1 expression, hence 0. Before the i-th operator the t-degree is $b - i + 1$ and the y_i-degree is r, so each operator is injective and $H_x = 0$.

- *Step 3 (induction).* For fixed x, t, the form $(Q_1, \dots, Q_r) \mapsto T(x^r Q_1, \dots, x^r Q_r, t^b)$ on $\operatorname{Sym}^m V$ has r slots and, by Step 2, vanishing diagonal on products. The induction hypothesis (which needs $m \geq r(r-1)$) makes it zero for *independent* $Q_1, \dots, Q_r$.
- *Step 4 (second differential sequence).* Start from $f = T(v_1^b, \dots, v_r^b, t^b)$ and apply D_{x,v_i} r times for each i. This trades r factors v_i of slot i for x^r, and the end result is a multiple of $T(x^r v_1^m, \dots, x^r v_r^m, t^b)$, which is 0 by Step 3. All r^2 operators raise the degree of the *same* variable x, so before each of them the x-degree is at most $r^2 - 1$ (it is 0 before the first and $r^2 - 1$ before the last); the v_i-degree is at least $m + 1$. Since $m + 1 > r^2 - 1$, each operator is injective and $f = 0$.

Where the bound comes from. The hypothesis enters only through $m \geq r^2$ (that is, $b \geq r(r+1) = a(a-1)$). What each step uses is this. Step 2 uses no lower bound on m: its degree gaps $m - i + 1$ $(1 \leq i \leq m)$ are automatically positive. Step 3 needs $m \geq 1$ and $m \geq r(r-1)$, in order to invoke the proposition for the pair (r, m). Step 4 needs $m + 1 > r^2 - 1$. All of these follow from $m \geq r^2$; in Step 4 the inequality holds with room to spare (the gap is at least 2). The quadratic size arises in Step 4, where r^2 derivatives accumulate on the one variable x, together with the inductive requirement of Step 3. The proof establishes sufficiency only; nothing here asserts that $b \geq a(a-1)$ is necessary.

Dependencies. Lemma 2 (powers span) ⇒ Lemma 3 (description of A_b) ⇒ Lemma 4 (criterion). Lemma 6 (injectivity) ⇒ Corollary 7 (chains of operators). Lemmas 2 and 8 (derivative rule) and Corollary 7 ⇒ Proposition 9. Lemma 4 and Proposition 9 ⇒ Theorem 1 ⇒ Corollary 10.

Changes of notation relative to the source. The source writes E both for $\operatorname{Sym}^b V$ and, inside the proof of the injectivity lemma, for the adjoint operator $D_{U,Y}$ of $D_{Y,U}$. Here $\operatorname{Sym}^b V$ is always written out, and that adjoint operator is called D'; it occurs only in the proof of Lemma 6. The operators applied in Proposition 9 are always written $D_{Y,U}$ with explicit blocks. The source's $P_a = (\operatorname{Sym} V)^{\otimes a}$ is written out, so P always denotes a product of b vectors. Statements left unnumbered in the source are now Lemmas 2, 3, 4, 8 and Corollary 7. The source's Lemma 2, Proposition 3, Corollary 4 and Remark 5 are Lemma 6, Proposition 9, Corollary 10 and Remark 11 here. The literature discussion of the source's introduction is collected in Section 6; no proof uses it.

2 The canonical image and diagonal vanishing

This section turns surjectivity of $\mu_{a,b,V}$ into a vanishing problem. An element $P = u_1 \cdots u_b \in \operatorname{Sym}^b V$ with $u_i \in V$ is called a *product of b vectors*; the vectors may repeat or be zero. Such products span $\operatorname{Sym}^b V$, because $\operatorname{Sym}^b V$ is the image of $V^{\otimes b}$.

Lemma 2 (Polarization; pure powers span)**.** *Let $k \geq 1$ and $u_1, \dots, u_k \in V$. In $\operatorname{Sym} V$,*

$$k!\, u_1 \cdots u_k = \sum_{J \subseteq \{1,\dots,k\}} (-1)^{k-|J|} \left(\sum_{j \in J} u_j \right)^k . \tag{3}$$

Consequently the pure powers u^k, $u \in V$, span $\operatorname{Sym}^k V$.

Used in: Lemma 3 (with $k = a$) and Step 5 of Proposition 9 (with $k = b$).

Proof. Step 1: the identity for formal variables. Work in $\mathbb{C}[z_1, \dots, z_k]$. By the multinomial theorem, $(\sum_{j \in J} z_j)^k = \sum_\beta \binom{k}{\beta} z^\beta$, the sum over exponent vectors β with $|\beta| = k$ whose support $S = \{j : \beta_j > 0\}$ lies in J. (For $J = \emptyset$ the left side is $0^k = 0$ because $k \geq 1$, and the sum over β is empty, so the formula holds there too.) So the coefficient of z^β on the right of (3) is

$$\binom{k}{\beta} \sum_{J \supseteq S} (-1)^{k-|J|} = \binom{k}{\beta} \sum_{i=0}^{s} \binom{s}{i} (-1)^{s-i} = \binom{k}{\beta} (1-1)^s, \qquad s = k - |S|,$$

where we wrote $J = S \sqcup J'$ and grouped by $i = |J'|$. This is 0 unless $s = 0$, i.e. $S = \{1, \dots, k\}$. With $|\beta| = k$ that forces $\beta = (1, \dots, 1)$, for which $\binom{k}{\beta} = k!$. So the right side equals $k!\, z_1 \cdots z_k$.

Step 2: substitution. $\operatorname{Sym} V$ is a commutative $\mathbb{C}$-algebra, so there is an algebra homomorphism $\mathbb{C}[z_1, \dots, z_k] \to \operatorname{Sym} V$ with $z_j \mapsto u_j$. Applying it to Step 1 gives (3).

Step 3: spanning. $\operatorname{Sym}^k V$ is spanned by products $u_1 \cdots u_k$. By (3) and $k! \neq 0$ in $\mathbb{C}$, each such product is a linear combination of k-th powers. □

Lemma 3 (The image)**.** *For integers $a, b \geq 1$,*

$$A_b = \operatorname{span}_{\mathbb{C}} \{P^{\otimes a} : P \text{ is a product of } b \text{ vectors}\} \subseteq R_b.$$

Used in: Lemma 4.

Proof. (1) For $u \in V$, formula (1) gives $\overline{\mathrm{Av}}(u^a) = u^{\otimes a} =: z(u) \in R_1$, since all $a!$ summands are equal. (2) By Lemma 2 with $k = a$, the u^a span $\operatorname{Sym}^a V$; since $\overline{\mathrm{Av}}$ is an isomorphism (Av3), the $z(u)$ span R_1. (3) A_b is spanned by products $z_1 \cdots z_b$ with $z_i \in R_1$, and the product is multilinear, so A_b is spanned by the products $z(u_1) \cdots z(u_b)$. (4) By the factorwise multiplication rule, $z(u_1) \cdots z(u_b) = (u_1 \cdots u_b)^{\otimes a}$. □

Under $R_b \cong \operatorname{Sym}^a(\operatorname{Sym}^b V)$ the tensor $P^{\otimes a}$ corresponds to the a-th power of P in $\operatorname{Sym}^a(\operatorname{Sym}^b V)$; this is the concrete formula for $\mu_{a,b,V}$ quoted in Section 1.

Lemma 4 (Diagonal criterion)**.** *Let $a, b \geq 1$. The map $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfying $T(P, \dots, P) = 0$ for all products P of b vectors is zero.*

Used in: the proof of Theorem 1 (only the direction "vanishing ⇒ surjective").

Proof. Recall $R_b = ((\operatorname{Sym}^b V)^{\otimes a})^{\mathfrak{S}_a}$, a subspace of $(\operatorname{Sym}^b V)^{\otimes a}$.

Step 1: forms on R_b are symmetric forms on $\operatorname{Sym}^b V$. For $\lambda \in R_b^*$ define

$$T_\lambda(w_1, \dots, w_a) = \lambda(\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)), \qquad w_i \in \operatorname{Sym}^b V.$$

This is a-linear, and symmetric by (Av2). We claim $\lambda \mapsto T_\lambda$ is a bijection from R_b^* onto the symmetric a-linear forms on $\operatorname{Sym}^b V$. *Injective:* if $T_\lambda = 0$ then λ vanishes on all $\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)$, which span R_b by (Av1). *Surjective:* given a symmetric a-linear T, let $\widetilde{T}\colon (\operatorname{Sym}^b V)^{\otimes a} \to \mathbb{C}$ be the linear map with $\widetilde{T}(w_1 \otimes \cdots \otimes w_a) = T(w_1, \dots, w_a)$ and put $\lambda = \widetilde{T}|_{R_b}$. Then

$$T_\lambda(w_1, \dots, w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} T(w_{\pi(1)}, \dots, w_{\pi(a)}) = T(w_1, \dots, w_a)$$

by symmetry of T.

Step 2: the diagonal. For a product P of b vectors, $\mathrm{Av}(P^{\otimes a}) = P^{\otimes a}$, so $T_\lambda(P, \dots, P) = \lambda(P^{\otimes a})$.

Step 3: conclusion. By Lemma 3 and Step 2, λ vanishes on A_b if and only if $T_\lambda(P,\dots,P)=0$ for all products P. A subspace A_b of the finite-dimensional space R_b equals R_b if and only if no nonzero $\lambda \in R_b^*$ vanishes on it. By Step 1 this says: $A_b = R_b$ if and only if the only symmetric T with vanishing diagonal on products is $T=0$. Finally $A_b = R_b$ is the surjectivity of $\mu_{a,b,V}$. □

Remark 5 (Reading of the criterion; not used later). A vector of V is a linear form on V^*, so a product $P = u_1 \cdots u_b$ is a degree-b form on V^* that splits into linear factors. The function $w \mapsto T(w,\dots,w)$ is a homogeneous polynomial of degree a on the vector space $\operatorname{Sym}^b V$, and the criterion tests it on split forms. So here b is the degree of the forms and a the degree of the test polynomial, whereas in the algebra A the integer a counts tensor factors and b is the grading degree. Lemma 4 is the duality between these two descriptions; compare the product parametrization in [11, Section 2].

3 Differential tools

The proof of Proposition 9 recovers a polynomial from a sequence of its directional derivatives. This section supplies the injectivity statement that lets each derivative be cancelled (Lemma 6, Corollary 7) and the rule for computing the derivatives (Lemma 8).

3.1 Vector variables

Assume $n = \dim V > 0$ and fix a basis $e_1,\dots,e_n$ of V. A *vector variable* U is a block of n independent indeterminates $U_1,\dots,U_n$. Let $\mathcal{P}$ be a polynomial ring over $\mathbb{C}$ in the coordinates of finitely many vector variables (and possibly further indeterminates). A polynomial is *homogeneous of degree p in U* if each of its monomials has total degree p in $U_1,\dots,U_n$; the zero polynomial is homogeneous of every degree. For two *different* blocks U, Y define the operator on $\mathcal{P}$

$$D_{Y,U} = \sum_{k=1}^{n} Y_k \frac{\partial}{\partial U_k}.$$

It lowers the U-degree by one, raises the Y-degree by one, and leaves the degree in every other block unchanged.

3.2 Injectivity

Lemma 6. *Let p, q be nonnegative integers. On polynomials homogeneous of degree p in U and degree q in Y, the operator $D_{Y,U}$ is injective if $p > q$. The polynomials may depend on any finite collection of additional variables independent of U and Y.*

Used in: Corollary 7, hence Steps 2 and 4 of Proposition 9.

The proof below is self-contained and is the only argument the paper relies on for this lemma. The source remarks that the statement is the positive-weight injectivity in $\mathfrak{sl}_2$ weight strings [6, Section 11.1, Claim 11.4 and equation (11.5)] and that Ikenmeyer [7, Section 5, Claim 5.1] uses a two-coordinate weight shift after decomposing tensors by untouched positions. These are comparisons, not ingredients.

Proof. Plan. With respect to a suitable inner product, the adjoint of $D = D_{Y,U}$ is $D' = D_{U,Y}$. The commutator $D'D - DD'$ acts on bihomogeneous polynomials as the scalar $p - q$. This gives the norm identity (8), which forbids $Dh = 0$ with $h \neq 0$ when $p > q$.

Step 1: the inner product. Let $X = (X_1, \dots, X_N)$ list all indeterminates of $\mathcal{P}$, including the additional ones. Define a Hermitian form (linear in the first argument, conjugate-linear in the second) by

$$\Big\langle \sum_\alpha c_\alpha X^\alpha, \sum_\beta d_\beta X^\beta \Big\rangle = \sum_\alpha \alpha!\, c_\alpha \overline{d_\alpha}, \qquad \alpha! := \prod_k \alpha_k!. \tag{4}$$

Distinct monomials are orthogonal, $\|X^\alpha\|^2 = \alpha!$, and $\langle g, h\rangle = \overline{\langle h, g\rangle}$. Since $\langle h, h\rangle = \sum_\alpha \alpha!\, |c_\alpha|^2$, the form is positive definite. This is the Fischer, or apolar, inner product [1, Section 2, equation (8) and Proposition 3].

Step 2: differentiation is adjoint to multiplication. For every indeterminate X_k and all $g, h \in \mathcal{P}$,

$$\langle \partial_{X_k} g, h\rangle = \langle g, X_k h\rangle. \tag{5}$$

Both sides are linear in g and conjugate-linear in h, so it suffices to take $g = X^\alpha$, $h = X^\beta$. The left side is $\alpha_k \langle X^{\alpha - e_k}, X^\beta\rangle$, which equals $\alpha_k (\alpha - e_k)! = \alpha!$ if $\alpha = \beta + e_k$ and 0 otherwise (including when $\alpha_k = 0$). The right side is $\langle X^\alpha, X^{\beta + e_k}\rangle$, the same number.

We also need the mirrored form. Apply (5) with the roles of g and h exchanged: $\langle \partial_{X_k} h, g\rangle = \langle h, X_k g\rangle$. Taking complex conjugates of both sides and using $\overline{\langle u, v\rangle} = \langle v, u\rangle$ gives

$$\langle g, \partial_{X_k} h\rangle = \langle X_k g, h\rangle. \tag{6}$$

Step 3: $D^* = D'$. Put $D' = D_{U,Y} = \sum_k U_k\, \partial_{Y_k}$. Then

$$\langle Dg, h\rangle = \sum_k \langle Y_k\, \partial_{U_k} g, h\rangle = \sum_k \langle \partial_{U_k} g, \partial_{Y_k} h\rangle = \sum_k \langle g, U_k\, \partial_{Y_k} h\rangle = \langle g, D' h\rangle,$$

where the second equality is (6) (for the indeterminate Y_k, with $\partial_{U_k} g$ in place of g) and the third is (5) (for U_k). Exchanging g and h and conjugating gives $\langle D' g, h\rangle = \langle g, Dh\rangle$.

Step 4: the commutator. Let $N_U = \sum_k U_k \partial_{U_k}$ and $N_Y = \sum_k Y_k \partial_{Y_k}$. As operators, $\partial_{Y_l} Y_k = \delta_{kl} + Y_k \partial_{Y_l}$ and $\partial_{U_k} U_l = \delta_{kl} + U_l \partial_{U_k}$, while U-operators commute with Y-operators. Hence

$$\begin{aligned} D'D &= \sum_{k,l} U_l\, \partial_{Y_l} Y_k\, \partial_{U_k} = N_U + \sum_{k,l} U_l Y_k\, \partial_{Y_l} \partial_{U_k}, \\ DD' &= \sum_{k,l} Y_k\, \partial_{U_k} U_l\, \partial_{Y_l} = N_Y + \sum_{k,l} Y_k U_l\, \partial_{U_k} \partial_{Y_l}, \end{aligned}$$

and the two double sums agree. Therefore

$$D'D - DD' = N_U - N_Y. \tag{7}$$

Step 5: the norm identity. Let h be homogeneous of degree p in U and q in Y. On a monomial, N_U multiplies by the total U-degree and N_Y by the total Y-degree (Euler), so $N_U h = ph$ and $N_Y h = qh$. By Step 3 and (7),

$$\|Dh\|^2 - \|D'h\|^2 = \langle h, D'Dh\rangle - \langle h, DD'h\rangle = \langle h, (N_U - N_Y)h\rangle = (p - q)\|h\|^2. \tag{8}$$

Step 6: conclusion. If $Dh = 0$ and $p > q$, then (8) reads $-\|D'h\|^2 = (p - q)\|h\|^2$. The left side is ≤ 0 and the right side is ≥ 0, so both vanish, and $p - q > 0$ gives $\|h\| = 0$, i.e. $h = 0$. □

The lemma applies to the *whole* space of polynomials of a given bidegree, with all other variables as spectators. So it can be applied repeatedly along a chain of operators, provided the degree inequality is checked before each one:

Corollary 7 (Cancelling a chain of derivatives)**.** *Let $F_0 \in \mathcal{P}$ and let $D_1, \dots, D_N$ be operators of the form $D_j = D_{Y^{(j)},U^{(j)}}$, where $U^{(j)} \neq Y^{(j)}$ are blocks of $\mathcal{P}$. Put $F_j = D_j F_{j-1}$. Suppose that for each j the polynomial F_{j-1} is homogeneous of some degree p_j in $U^{(j)}$ and some degree q_j in $Y^{(j)}$, with $p_j > q_j$. If $F_N = 0$, then $F_0 = 0$.*

Proof. Descending induction on j: if $F_j = 0$, then $D_j F_{j-1} = 0$, and Lemma 6 (with $p = p_j$, $q = q_j$) gives $F_{j-1} = 0$. □

3.3 Evaluating a multilinear form on vector variables

We make precise what an expression such as $T(x^r Q, \dots, x^r Q, t^b)$ means when t and the factors of Q are vector variables, and how $D_{Y,U}$ acts on it.

Conventions. Elements of $\mathcal{P} \otimes_{\mathbb{C}} \operatorname{Sym} V$ are polynomials in the indeterminates with coefficients in $\operatorname{Sym} V$; this is a commutative ring. A vector variable U is identified with the element $\sum_k U_k e_k$. An a-linear form T on $\operatorname{Sym}^b V$ extends uniquely to a $\mathcal{P}$-multilinear map $(\mathcal{P} \otimes \operatorname{Sym}^b V)^a \to \mathcal{P}$:

$$T(f_1 s_1, \dots, f_a s_a) = f_1 \cdots f_a \, T(s_1, \dots, s_a), \qquad f_i \in \mathcal{P},\ s_i \in \operatorname{Sym}^b V.$$

So T of arguments involving vector variables is an element of $\mathcal{P}$, an ordinary polynomial with complex coefficients.

Evaluation. Assigning complex values to all indeterminates is a ring homomorphism $\mathrm{ev}\colon \mathcal{P} \to \mathbb{C}$; applied to coefficients it gives a ring homomorphism $\mathcal{P} \otimes \operatorname{Sym} V \to \operatorname{Sym} V$, also written ev. It sends a vector variable $U = \sum_k U_k e_k$ to the vector $\sum_k \mathrm{ev}(U_k) e_k$, and every vector of V arises for a suitable choice of values. It is compatible with T:

$$\mathrm{ev}\big(T(g_1, \dots, g_a)\big) = T(\mathrm{ev}\, g_1, \dots, \mathrm{ev}\, g_a).$$

Both sides are additive in each g_i, and for $g_i = f_i s_i$ both equal $\mathrm{ev}(f_1) \cdots \mathrm{ev}(f_a)\, T(s_1, \dots, s_a)$, the right side by multilinearity of T.

Zero principle. A polynomial $F \in \mathbb{C}[X_1, \dots, X_N]$ that vanishes at every point of $\mathbb{C}^N$ is zero. Induction on N: write $F = \sum_j c_j X_N^j$ with $c_j \in \mathbb{C}[X_1, \dots, X_{N-1}]$. For fixed values of $X_1, \dots, X_{N-1}$ the resulting polynomial in X_N has infinitely many roots, so all its coefficients vanish; thus each c_j vanishes at every point of $\mathbb{C}^{N-1}$ and is zero by induction. Combined with evaluation: an expression $T(\cdots) \in \mathcal{P}$ in vector variables is the zero polynomial as soon as it vanishes for all choices of actual vectors.

Derivatives. $D_{Y,U}$ acts on $\mathcal{P} \otimes \operatorname{Sym} V$ through the coefficients: $D_{Y,U}(fs) = D_{Y,U}(f)\, s$. An element is called *$U$-free* if its coefficients do not involve $U_1, \dots, U_n$.

Lemma 8 (Derivative rule)**.** *Let $U \neq Y$ be blocks of $\mathcal{P}$, $D = D_{Y,U}$, and T an a-linear form on $\operatorname{Sym}^b V$. Let $1 \le j \le b$, let $w \in \mathcal{P} \otimes \operatorname{Sym}^{b-j} V$ and $g_1, \dots, g_{a-1} \in \mathcal{P} \otimes \operatorname{Sym}^b V$ all be U-free. Then*

$$D\,T(g_1, \dots, g_{a-1},\, U^j w) = j\, T(g_1, \dots, g_{a-1},\, U^{j-1} Y w). \tag{9}$$

The same identity holds, with the same proof, when $U^j w$ stands in any other slot and the remaining $a-1$ arguments are U-free; symmetry of T is not needed. The elements w, g_i may involve Y.

Used in: formulas (14) and (17).

Proof. (a) *Leibniz rules.* For $g, h \in \mathcal{P} \otimes \operatorname{Sym} V$ we have $D(gh) = D(g)h + gD(h)$: both sides are biadditive, and for $g = fs$, $h = f's'$ the claim is $D(ff')ss' = (D(f)f' + fD(f'))ss'$, the Leibniz rule in $\mathcal{P}$. Similarly $D\,T(g_1, \dots, g_a) = \sum_i T(g_1, \dots, Dg_i, \dots, g_a)$: both sides are additive in each g_i, and for $g_i = f_i s_i$ the left side is $D(f_1 \cdots f_a)\,T(s_1, \dots, s_a)$ with $T(s_1, \dots, s_a) \in \mathbb{C}$ a constant, which the Leibniz rule in $\mathcal{P}$ expands into the right side.

(b) $D(U) = \sum_k D(U_k)e_k = \sum_k Y_k e_k = Y$, and $D(g) = 0$ for U-free g, because D only differentiates in $U_1, \dots, U_n$.

(c) By (a) and (b), $D(U^j) = jU^{j-1}Y$ and $D(U^j w) = jU^{j-1}Yw + U^j D(w) = jU^{j-1}Yw$. In the Leibniz rule for T every term in which D falls on a U-free argument vanishes by (b), whichever slot holds $U^j w$. This leaves (9). □

4 Vanishing on products

Proposition 9. *Let V be a finite-dimensional complex vector space, and let $a \geq 1$ and $b \geq 1$ be integers satisfying $b \geq a(a-1)$. If a symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfies*

$$T(P, \dots, P) = 0 \qquad \text{for every product } P \text{ of } b \text{ vectors}, \tag{10}$$

then $T = 0$.

Used in: the proof of Theorem 1, via Lemma 4.

Proof. Plan. Induction on a, for all admissible b and all V at once. In the inductive step we show $T(v_1^b, \dots, v_{a-1}^b, t^b) = 0$ for all vectors, via Steps 1–4 of Section 1.3; Step 5 concludes by spanning.

Trivial case and base case. If $V = 0$ then $\operatorname{Sym}^b V = 0$ (as $b \geq 1$) and $T = 0$. So let $n = \dim V > 0$ and fix a basis as in Section 3. For $a = 1$, T is a linear functional vanishing on all products of b vectors; these span $\operatorname{Sym}^b V$ (start of Section 2), so $T = 0$.

Inductive setup. Let $a \geq 2$ and assume the proposition for $a - 1$ (all admissible degrees, all V). Put

$$r = a - 1 \geq 1, \qquad m = b - r. \tag{11}$$

From $b \geq a(a-1) = r(r+1)$ we get

$$b \geq r(r+1), \qquad m \geq r(r+1) - r = r^2 \geq 1. \tag{12}$$

Step 1: a split binary form. *Claim.* For all $x, t \in V$ and every product $Q = y_1 \cdots y_m$ of m vectors,

$$T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^r Q) = 0. \tag{13}$$

(1a) *$X^r + cY^r$ splits.* Let $c \in \mathbb{C}$. If $c = 0$ the polynomial is X^r. If $c \neq 0$, choose $\alpha \in \mathbb{C}$ with $\alpha^r = -c$ and let $\zeta = e^{2\pi i/r}$. In $\mathbb{C}[X, Y]$,

$$X^r + cY^r = X^r - \alpha^r Y^r = \prod_{j=0}^{r-1} (X - \alpha\zeta^j Y),$$

because $X^r - \alpha^r Y^r$ is monic of degree r in X over the domain $\mathbb{C}[Y]$ and has the r distinct roots $X = \alpha\zeta^j Y$.

(1b) *Substitution.* For any $x, t \in V$ (no independence needed) there is an algebra homomorphism $\mathbb{C}[X, Y] \to \operatorname{Sym} V$ with $X \mapsto x$, $Y \mapsto t$, because $\operatorname{Sym} V$ is commutative. It maps the factor $X - \alpha\zeta^j Y$ to the *vector* $x - \alpha\zeta^j t \in V$. Hence $x^r + ct^r$ is a product of r vectors, and

$$P_c := (x^r + ct^r)Q = x^r Q + c\, t^r Q$$

is a product of $r + m = b$ vectors, for every $c \in \mathbb{C}$.

(1c) *Coefficient of c.* By (10), $T(P_c, \dots, P_c) = 0$ for all $c \in \mathbb{C}$. Expanding by multilinearity, $T(P_c, \dots, P_c)$ is a polynomial in c of degree at most a. Its coefficient of c^1 is the sum of the a terms with $t^r Q$ in exactly one slot and $x^r Q$ in the others; by symmetry of T these are equal, giving $a\, T(x^r Q, \dots, x^r Q, t^r Q)$. A polynomial in c vanishing for all $c \in \mathbb{C}$ has all coefficients zero, and $a \neq 0$. This proves (13).

Step 2: freeing the last slot. *Claim.* For all $x, t \in V$ and every product Q of m vectors, $T(x^r Q, \dots, x^r Q,\, t^b) = 0$.

Fix $x \in V$. Let $\mathcal{P}$ be the polynomial ring in the coordinates of independent vector variables $t, y_1, \dots, y_m$, put $Q = y_1 \cdots y_m$, and, for $0 \le k \le m$,

$$G_k = \frac{b!}{(b-k)!}\, T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}},\, t^{b-k} y_1 \cdots y_k) \in \mathcal{P}, \qquad H_x := G_0 = T(x^r Q, \dots, x^r Q,\, t^b).$$

(2a) $G_k = D_{y_k,t} G_{k-1}$ *for* $1 \le k \le m$, that is,

$$D_{y_k,t} \cdots D_{y_1,t} H_x = \frac{b!}{(b-k)!}\, T(x^r Q, \dots, x^r Q,\, t^{b-k} y_1 \cdots y_k). \tag{14}$$

Apply Lemma 8 with $U = t$, $Y = y_k$, $j = b - k + 1$ and $w = y_1 \cdots y_{k-1}$. The first r arguments $x^r Q$ and w are t-free, so only the last slot is differentiated, and the scalar is $\frac{b!}{(b-k+1)!}(b - k + 1) = \frac{b!}{(b-k)!}$.

(2b) $G_m = 0$. Since $b - m = r$, we have $G_m = \frac{b!}{r!} T(x^r Q, \dots, x^r Q,\, t^r Q)$. At every assignment of actual vectors to $t, y_1, \dots, y_m$ this vanishes by (13), so $G_m = 0$ in $\mathcal{P}$ by the zero principle.

(2c) *Degrees before the i-th operator.* Let $1 \le i \le m$. Under $\mathcal{P}$-multilinearity the degrees of the a arguments add. Expanding $y_l = \sum_k y_{l,k} e_k$ shows that each of the r arguments $x^r Q$ has degree 0 in t and degree 1 in y_i (the factor y_i occurs in Q exactly once). The last argument $t^{b-i+1} y_1 \cdots y_{i-1}$ has degree $b - i + 1$ in t and degree 0 in y_i (it contains only $y_1, \dots, y_{i-1}$). Hence G_{i-1} is homogeneous

$$\text{of degree } p = r \cdot 0 + (b - i + 1) = b - i + 1 \text{ in } t \quad \text{and of degree } q = r \cdot 1 + 0 = r \text{ in } y_i,$$

and $p - q = m - i + 1 \ge 1$. The earlier operators $D_{y_1,t}, \dots, D_{y_{i-1},t}$ raised the degrees of $y_1, \dots, y_{i-1}$, not of y_i; this is why $q = r$ and not $r + 1$.

(2d) *Conclusion.* By (2a)–(2c), Corollary 7 applies to $F_j = G_j$, $D_j = D_{y_j,t}$, $N = m$, and gives $H_x = G_0 = 0$ in $\mathcal{P}$. Evaluating at actual vectors proves the claim.

Step 3: induction on the remaining slots. *Claim.* For all $x, t \in V$,

$$T(x^r Q_1, \dots, x^r Q_r,\, t^b) = 0 \qquad \text{for all } Q_1, \dots, Q_r \in \operatorname{Sym}^m V. \tag{15}$$

Fix $x, t \in V$ and define $S_{x,t}(Q_1, \dots, Q_r) := T(x^r Q_1, \dots, x^r Q_r, t^b)$. Multiplication by x^r is a linear map $\operatorname{Sym}^m V \to \operatorname{Sym}^b V$ and T is symmetric and multilinear, so $S_{x,t}$ is a symmetric r-linear form on $\operatorname{Sym}^m V$. By Step 2, $S_{x,t}(Q, \dots, Q) = 0$ for every product Q of m vectors. The induction hypothesis applies to $S_{x,t}$ with (a, b) replaced by (r, m): indeed $r \ge 1$, $m \ge 1$, and $m \ge r^2 \ge r(r-1)$ by (12). Hence $S_{x,t} = 0$.

Step 4: removing the common power. *Claim.* $T(v_1^b,\dots,v_r^b,t^b)=0$ for all $v_1,\dots,v_r,t\in V$.

In (15) the Q_i are independent; the only remaining constraint is the common factor x^r. Fix $t\in V$. Let $\mathcal{P}$ now be the polynomial ring in the coordinates of independent vector variables $v_1,\dots,v_r$ and one further vector variable x (in Steps 1–3, x was a vector; here it is a variable). Put

$$f=T(v_1^b,\dots,v_r^b,t^b)\in\mathcal{P},$$

which has x-degree 0. Apply r^2 operators in the following order: D_{x,v_1} r times, then D_{x,v_2} r times, …, then D_{x,v_r} r times. For $1\le i\le r$ and $0\le s\le r$ let $f_{i,s}$ be the result after all operators for $v_1,\dots,v_{i-1}$ and s operators for v_i (so $f_{1,0}=f$ and $f_{i,r}=f_{i+1,0}$ for $i<r$).

(4a) *Formula.* Recall $b-r=m$. For all such i,s,

$$f_{i,s}=\left(\frac{b!}{m!}\right)^{i-1}\frac{b!}{(b-s)!}\,T\big(x^rv_1^m,\dots,x^rv_{i-1}^m,\ x^sv_i^{b-s},\ v_{i+1}^b,\dots,v_r^b,\ t^b\big).\tag{16}$$

We prove (16) by induction along the order in which the operators are applied, i.e. on (i,s) in lexicographic order.

- *Start,* $(i,s)=(1,0)$. The right side of (16) is $1\cdot\frac{b!}{b!}\,T(v_1^b,\dots,v_r^b,t^b)=f$.
- *From* (i,s) *to* $(i,s+1)$*, for* $0\le s\le r-1$. Here $f_{i,s+1}=D_{x,v_i}f_{i,s}$. Apply Lemma 8 (in its "any slot" form, for slot i) with $U=v_i$, $Y=x$, $j=b-s\ge m+1\ge 1$ and $w=x^s\in\mathcal{P}\otimes\mathrm{Sym}^s V$. The cofactor x^s and the other slots $x^rv_l^m$ $(l<i)$, v_l^b $(l>i)$, t^b are v_i-free: they involve only x, the blocks v_l with $l\ne i$, and the constant t. So only slot i changes, from $x^sv_i^{b-s}$ to $(b-s)\,x^{s+1}v_i^{b-s-1}$, and the scalar becomes $\frac{b!}{(b-s)!}(b-s)=\frac{b!}{(b-s-1)!}$. This is (16) for $(i,s+1)$. In particular the previously inserted factors x are not differentiated.
- *From* (i,r) *to* $(i+1,0)$*, for* $i<r$. These are the same polynomial, and the two instances of (16) agree: for (i,r), slot i is $x^rv_i^{b-r}=x^rv_i^m$ with scalar $(b!/m!)^{i-1}\cdot b!/(b-r)!=(b!/m!)^i$; for $(i+1,0)$, slot $i+1$ is $x^0v_{i+1}^b$ with scalar $(b!/m!)^i\cdot b!/b!$.

At the end, $(i,s)=(r,r)$ gives

$$D_{x,v_r}^r\cdots D_{x,v_1}^r f=f_{r,r}=\left(\frac{b!}{m!}\right)^r T(x^rv_1^m,\dots,x^rv_r^m,t^b)=0.\tag{17}$$

The last equality holds at every assignment of actual vectors by (15) with $Q_i=v_i^m$, hence in $\mathcal{P}$ by the zero principle.

(4b) *Degrees before each operator.* The operator applied to $f_{i,s}$ with $0\le s\le r-1$ is D_{x,v_i}. Reading off (16) (degrees of the arguments add): v_i occurs only in slot i, with degree $b-s$; and x occurs with degree r in each of the $i-1$ completed slots and degree s in slot i. So $f_{i,s}$ is homogeneous of degree $p=b-s$ in v_i and of degree $q=(i-1)r+s$ in x. Since $s\le r-1$ and $i\le r$,

$$p\ge b-r+1=m+1,\qquad q\le(r-1)r+(r-1)=r^2-1,\qquad p-q\ge m-r^2+2\ge 2>0,$$

using $m\ge r^2$ from (12). (Only $p-q>0$ is needed.) For example, for $r=2$ the four operators see $(p,q)=(b,0),(b-1,1),(b,2),(b-1,3)$.

(4c) *Conclusion.* By (4a), (4b) and (17), Corollary 7 applies to the chain of r^2 operators and gives $f=0$ in $\mathcal{P}$. Evaluating at actual vectors, and recalling that $t\in V$ was arbitrary, proves the claim.

Step 5: conclusion. By Lemma 2 with $k = b$, the pure powers v^b span $\mathrm{Sym}^b V$. By Step 4 (recall $a = r + 1$), T vanishes whenever all a arguments are pure powers; by multilinearity $T = 0$. This completes the induction. □

Sanity check: $a = 2$. Here $r = 1$, $m = b - 1 \geq 1$, and the bound is $b \geq 2$. Step 1 gives $T(xQ, tQ) = 0$. Step 2 applies $b - 1$ operators to $T(xQ, t^b)$ and reaches $b!\,T(xQ, tQ) = 0$, with degree gaps $b - 1, b - 2, \ldots, 1$. Step 3 is the base case: $Q_1 \mapsto T(xQ_1, t^b)$ is linear and vanishes on products. Step 4 is the single operator D_{x,v_1} on $T(v_1^b, t^b)$, which gives $b\,T(xv_1^{b-1}, t^b) = 0$; it is injective since $b > 0$.

5 Proof of the theorem and a corollary

Proof of Theorem 1. Let $a \geq 2$ and $b \geq a(a-1)$; then $b \geq 2$, so $a, b \geq 1$. By Proposition 9, every symmetric a-linear form on $\mathrm{Sym}^b V$ whose diagonal vanishes on products of b vectors is zero. By Lemma 4, $\mu_{a,b,V}$ is surjective. □

Corollary 10. *For every finite-dimensional complex vector space V, every integer $a \geq 2$, and every integer $b \geq a(a-1)$, there exists a $\mathrm{GL}(V)$-equivariant embedding*

$$\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V).$$

In particular, taking $a = 6$ (so $a(a-1) = 30$), there is such an embedding $\mathrm{Sym}^6(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^6 V)$ for every $b \geq 30$; this is what the source calls "the sixth symmetric-power comparison for $b \geq 30$".

Proof. (1) *$\mu_{a,b,V}$ is equivariant.* $\mathrm{GL}(V)$ acts on $\mathrm{Sym}\, V$ by algebra automorphisms, $g \cdot (u_1 \cdots u_j) = (gu_1) \cdots (gu_j)$, hence on each $W = \mathrm{Sym}^j V$ and diagonally on $W^{\otimes a}$. The diagonal action commutes with permuting tensor factors, so it preserves each R_j and, by (1), commutes with Av. Therefore $\overline{\mathrm{Av}}\colon \mathrm{Sym}^a W \to (W^{\otimes a})^{\mathfrak{S}_a}$ is equivariant, and so are the identifications $R_1 \cong \mathrm{Sym}^a V$ and $R_b \cong \mathrm{Sym}^a(\mathrm{Sym}^b V)$. The diagonal action also respects factorwise multiplication in $(\mathrm{Sym}\, V)^{\otimes a}$, so the multiplication map $\mathrm{Sym}^b R_1 \to R_b$ is equivariant. Hence $\mu_{a,b,V}$ is equivariant; it is surjective by Theorem 1. (2) Finite-dimensional polynomial representations of $\mathrm{GL}(V)$ over $\mathbb{C}$ are completely reducible [9, Chapter I, Appendix A, Sections 7–8]. So $\ker \mu_{a,b,V}$ has an invariant complement M in $\mathrm{Sym}^b(\mathrm{Sym}^a V)$. (3) The restriction $\mu_{a,b,V}|_M$ is an equivariant isomorphism onto $\mathrm{Sym}^a(\mathrm{Sym}^b V)$. Its inverse followed by the inclusion $M \subseteq \mathrm{Sym}^b(\mathrm{Sym}^a V)$ is the required embedding. This shows existence only; the embedding is not identified with any prescribed canonical map. □

Remark 11 (Scope). This remark restates Theorem 1; it claims nothing beyond it (in particular nothing about infinite-dimensional V). Lemma 6 holds for any number n of coordinates and any spectator variables, and the degree comparisons in Proposition 9 involve only a and b. Together with the finite-dimensional duality of Lemma 4, this is why the same bound holds for every finite-dimensional V. The argument treats each V directly: it does not first prove the statement in one convenient dimension and then transfer it (the source's "stable-dimension reduction"). The source names three features of $\mathbb{C}$ used by the written proof: positivity of the Fischer Hermitian inner product (Lemma 6); splitting of binary forms, which rests on r-th roots (Step 1); and invertibility of the integer factors $k!$, a, $b!/(b-k)!$ and $(b!/m!)^r$ in the polarization, coefficient and differentiation identities. This list is not claimed to be exhaustive; for instance, the zero principle also uses that $\mathbb{C}$ is infinite.

6 Context from the literature (not used in the proofs)

Everything in this section is reported from the source with its citations. None of it was checked for this rewrite, and the source does not reproduce the cited statements; read each sentence as "the source states that ...". Nothing in Sections 2–5 depends on this section.

The map in the literature. In positive dimension, Raicu–Sam–Weyman [11, Section 2] formulate the map as $\mathrm{Sym}^b(D^aV) \longrightarrow D^a(\mathrm{Sym}^b V)$, with divided powers realized as invariant tensors. Transporting both sides by the averaging isomorphism (Av3) gives (2).

Geometry. The affine Chow variety of decomposable a-forms consists of products of a linear forms in $\mathrm{Sym}^a(V^*)$. Its coordinate ring identifies with A, whose normalization is R; here b is the coordinate degree [11, Section 2, Theorem 2.3]. See also Landsberg [8, Section 7.6, Propositions 7.11–7.12].

Stabilization. For fixed a and V, a *stabilization degree* is an integer $b_0 \geq 0$ such that $\mu_{a,b,V}$ is surjective for every $b \geq b_0$. One may ask for the least such b_0 in a fixed dimension, or for a bound valid in all finite dimensions at once; Theorem 1 is of the second kind. Brion established eventual surjectivity [2] and then gave an effective bound involving both a and the dimension [3, Theorem 3.3]. Since $(R/A)_b = R_b/A_b$ is nonzero exactly when $\mu_{a,b,V}$ is not surjective, eventual surjectivity says that R/A has only finitely many nonzero graded components, and the least stabilization degree is one more than the highest nonzero degree of R/A, or zero if $R = A$. Raicu, Sam, and Weyman bound the regularity of the Segre ring and modules of covariants, including the normalization [11, Theorem 3.9]; according to the source, their theorem does not compute the top degree of the quotient R/A. The source also states that Theorem 1 answers positively the request for a polynomial bound in [8, Problem 7.19]: in that reference's notation $d = b$, $n = a$ and $w = \dim V$, and the bound reads $d \geq n(n-1)$, a polynomial in n not involving w, valid in particular when $w = n$. The problem's wording is not reproduced in the source, so neither the translation of notation nor the claim that the problem is answered can be checked from the manuscript; both are reported, not verified. The present argument does not determine the least stabilization degree.

Relation to the Foulkes conjecture. The ordinary Foulkes conjecture [5] asks, for $a \leq b$, for a $\mathrm{GL}(V)$-equivariant injection $\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$ [11, Conjecture 2.10]. Corollary 10 provides one in the range $b \geq a(a-1)$. The conjecture is about the existence of some embedding, whereas Theorem 1 is about one specified map. The difference is visible at $a = b$: source and target of $\mu_{a,a,V}$ are the same representation, so an embedding exists trivially, yet $\mu_{a,a,V}$ need not be invertible. Since source and target have equal dimension, a nonzero kernel is the same as a failure of surjectivity. The source reports that the canonical map has nonzero kernel at $(a,b) = (5,5)$ and $(6,6)$ in suitable dimensions [4, arXiv version, Theorem 6(b),(c) and Remark 7]; it does not say which dimensions, and the claim was not checked here. Whatever the dimensions are, there is no conflict with Theorem 1: the theorem covers every V, but for $a = 5$ only $b \geq 20$ and for $a = 6$ only $b \geq 30$.

Relation of the method to earlier work. Successive weight shifts also occur in Ikenmeyer's proof [7, Section 5] of McKay's propagation theorem [10]. That setting differs from ours in two ways. It concerns the canonical maps in the *opposite* direction $\mathrm{Sym}^a(\mathrm{Sym}^b V) \longrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$, and it assumes $\dim V \geq b$ for each b. The result is a propagation statement: for fixed a and $B \geq a$, injectivity at (a,B) propagates to every (a,b) with $b \geq B$ [7, Theorem 1.3 and Section 3]. Here there

is no dimension hypothesis; the two differential sequences (Steps 2 and 4) surround an induction on the number of slots (Step 3) and give a surjectivity range directly.

7 Verification checklist

No step in Sections 2–5 carries an "unverified" mark, because no step was found that could not be justified. That is a report, not a proof: the items below are the places where an error would do the most damage and should be checked first. The literature claims of Section 6 were not checked and are not used.

1. **Duality** (Lemma 4, Steps 1–3): that $\lambda \mapsto T_\lambda$ is a bijection, and that A_b is spanned by the tensors $P^{\otimes a}$ (Lemma 3, which uses Lemma 2 with $k = a$).

2. **Injectivity lemma** (Lemma 6): the adjointness (5) and its mirrored form (6), the sign in the commutator (7), and the strict inequality $p > q$ in Step 6. Also that U and Y must be different blocks.

3. **Step 1** of Proposition 9: that $(x^r + ct^r)Q$ really is a product of b vectors for *every* c ((1a)–(1b), using roots in $\mathbb{C}$), and that the coefficient of c is $a\,T(x^rQ, \dots, x^rQ, t^rQ)$ (1c).

4. **Step 2 degree count** (2c): before the i-th operator the t-degree is $b - i + 1$ and the y_i-degree is r, since y_i occurs once in each of the r slots x^rQ and not yet in the last slot; so $p - q = m - i + 1 \geq 1$.

5. **Step 3**: the induction hypothesis is applied to an r-linear form on $\mathrm{Sym}^m V$, which requires $m \geq 1$ and $m \geq r(r-1)$; both follow from (12).

6. **Step 4 formula and degree count** ((4a), (4b)): the induction proving (16), including the hand-over from (i, r) to $(i+1, 0)$; then that all r^2 operators raise the degree of the same block x, so $q = (i-1)r + s$ can be as large as $r^2 - 1$, while $p = b - s \geq m + 1$; the inequality $m + 1 > r^2 - 1$ follows from $m \geq r^2$.

7. **Derivative rule** (Lemma 8): it requires only that the other slots and the cofactor w be free of the differentiated variable U; they may contain Y, and the slot holding $U^j w$ is arbitrary. This is what justifies (14) and (17).

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